

Human-Machine Collaboration at the Tollbooth: Perceptions and Requirements of Toll Workers in the Age of AI and IoT

Nurayad Nafiz Islam

Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
nafizislam09@gmail.com

A. B. M. Alim Al Islam

Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
alim_razi@cse.buet.ac.bd

Abstract

In the context of developing countries like Bangladesh, the advancement of toll collection systems holds significant implications for transportation infrastructure and economic development. This paper investigates the perspectives of toll collectors in Bangladesh towards the integration of Artificial Intelligence (AI) and Internet of Things (IoT) technologies, as well as their requirements and perceptions concerning toll collection software. Through a mixed-methods approach involving interviews and surveys, insights were gathered from toll collectors operating within the Bangladeshi toll booth ecosystem. Results illustrate a nuanced spectrum of attitudes towards AI and IoT adoption, reflecting both cautious optimism and skepticism amidst concerns about job security and technology literacy. Additionally, toll collectors articulated essential requirements such as user-friendly interfaces, robustness in low-resource settings, and compatibility with local languages and operational practices. Perceptions of toll collection software underscored the importance of affordability, reliability, and adaptability to the unique challenges of Bangladeshi toll collection environments. By illuminating toll collectors' perspectives within a developing country context, this study offers valuable insights to inform the design and implementation of AI, IoT, and software solutions tailored to the needs and constraints of Bangladesh's tolling infrastructure, thereby facilitating more inclusive and effective toll collection operations in the region.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Field studies**; **Interaction devices**;

Keywords

ITS, HCI, AI, IOT

ACM Reference Format:

Nurayad Nafiz Islam and A. B. M. Alim Al Islam. 2024. Human-Machine Collaboration at the Tollbooth: Perceptions and Requirements of Toll Workers in the Age of AI and IoT. In *11th International Conference on Networking, Systems, and Security (NSysS '24)*, December 19–21, 2024, Khulna, Bangladesh. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3704522.3704545>



This work is licensed under a Creative Commons Attribution International 4.0 License.

NSysS '24, December 19–21, 2024, Khulna, Bangladesh
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1158-9/24/12
<https://doi.org/10.1145/3704522.3704545>

1 Introduction

Traffic congestion is a persistent challenge in developing countries, impacting economic productivity and citizen well-being. In response, governments and transportation authorities are increasingly turning to technological advancements to improve efficiency and manage traffic flow. One key area targeted for improvement is toll collection, where manual processes are being replaced with automated solutions utilizing Artificial Intelligence (AI) and Internet of Things (IoT) devices. While these advancements hold the promise of smoother traffic flow and faster revenue collection, the human impact of such transformations deserves closer scrutiny, particularly in the context of developing nations.

This study delves into the human-centered aspects of AI and IoT adoption in Bangladesh's toll collection systems. Bangladesh, a nation experiencing rapid economic growth and urbanization, provides a compelling case study for such research. Here, the implementation of new technologies coexists with a diverse populace with varying levels of technological literacy. We focus on toll collectors, often overlooked yet crucial players in the transportation ecosystem. Their experiences with these new systems hold valuable insights into the effectiveness and potential downsides of automation in toll collection.

Our research investigates two key questions that address the potential drawbacks alongside the promised benefits of automation:

RQ1: Do AI and IoT based toll collection systems truly assist the transportation system, or do they create unforeseen complexities? While automation aims to streamline traffic flow, concerns exist that it might introduce new challenges for booth operations. This study investigates the experiences of toll collectors to understand whether these systems simplify their work or create unexpected complexities that could hinder smooth traffic flow. For example, are the systems intuitive for them to use? Do they encounter technical difficulties or require additional training to operate these new systems effectively?

RQ2: Are these AI and IoT powered toll collection systems easily accessible for people with varying levels of education? Inclusivity is a critical aspect of implementing new technologies. We examine user perspectives to determine if the systems are readily understandable and usable by individuals with diverse educational backgrounds. Is the user interface clear and intuitive? Are there language barriers for non-tech-savvy users? Do some user groups encounter difficulties interacting with the automated systems?

By investigating these questions, we aim to bridge the gap between technological advancements and the human element of toll collection. Understanding the needs and experiences of toll collectors and users in a developing country context is crucial for designing efficient and user-centered toll collection systems that are accessible to all. This research can inform policy decisions and guide the development of future technologies that optimize the transportation system while considering the well-being and capabilities of people who interact with them.

2 Motivation

Recent developments in AI and IoT have rapidly transformed various sectors, including transportation and infrastructure management. These technologies offer innovative solutions for automating toll collection, optimizing traffic flow, and enhancing operational efficiency. This research explores how AI and IoT technologies can be strategically implemented in the toll collection industry to address existing inefficiencies while also being mindful of their broader socio-economic implications.

2.1 Research Context



Figure 1: A Toll Operator

In the contemporary landscape of toll collection systems, the convergence of Artificial Intelligence (AI) and Internet of Things (IoT) technologies represents a paradigm shift in transportation infrastructure management. While these advancements hold promise for streamlining toll collection processes, particularly in developing countries like Bangladesh, their implementation raises pertinent questions about their socio-economic impact and usability.

The context of Bangladesh, with its evolving transportation infrastructure and diverse socio-economic landscape, presents a compelling backdrop for exploring the implications of AI, IoT, and toll collection software in real-world settings. Understanding the perspectives of toll collectors within this context is essential for unraveling the complexities and nuances of technology adoption in toll collection operations.



Figure 2: A Toll Plaza in Bangladesh

2.2 Applications

The insights generated from this research have multifaceted applications across various domains, including policy formulation, technology development, and infrastructure management. Firstly, policymakers and government agencies can utilize the findings to devise informed strategies for integrating AI, IoT, and toll collection software into the transportation infrastructure of Bangladesh. By aligning technology initiatives with the needs and realities of toll collectors, policymakers can foster more inclusive and sustainable development in the tolling sector. Additionally, technology developers and solution providers stand to benefit from understanding the requirements and challenges faced by toll collectors, enabling them to tailor their offerings to suit the local context effectively. Moreover, the outcomes of this research can inform capacity-building initiatives aimed at enhancing the digital literacy and skillsets of toll collectors, thereby fostering greater participation and empowerment within the workforce. Overall, the applications of this work extend beyond academic discourse, offering actionable insights to stakeholders involved in shaping the future of toll collection systems in Bangladesh and similar contexts.

3 Related Work

This research topic suffers from a significant lack of study and investigation, creating a vacuum of knowledge and insight in the field. Prioritizing this area of research will be crucial to fill the gap and advance our understanding.

One similar work is [10] where the authors discuss the significance of Electronic Toll Collection (ETC) systems in the context of developing countries like Bangladesh. It highlights the increasing pressure on highways due to the growing number of vehicles and the implementation of Intelligent Transportation Systems (ITS) as a solution. Several countries have already adopted ETC technologies, such as Singapore, Japan, Germany, Hong Kong, Stockholm, China, and America, each utilizing different systems like RFID, GNSS, DSRC, and AV-CV technology.

In Bangladesh, the implementation of ETC faces infrastructure challenges, but it is seen as a solution for better traffic management, reduced CO₂ emissions, and increased road lifespan. The paper

suggests that RFID-based ETC could be a cost-effective short-term solution for Bangladesh. A survey indicates that 89% of the population believes ETC can alleviate highway traffic congestion. The Padma Bridge and Dhaka-Mawa-Bhanga expressway are identified as locations for introducing automated toll collection.

In conclusion, the paper emphasizes the importance of implementing ETC systems in developing countries like Bangladesh to address the increasing number of vehicles and improve traffic management. It suggests considering technologies like Satellite-based Electronic Road Pricing System for long-term benefits. The paper calls for proactive steps from the authorities to gradually implement ETC systems across the country to support socioeconomic development over the next decade.

Another work is [2] where the authors compare conventional toll collection methods with automatic toll collection systems like FASTag and BookMyToll. It discusses the increasing traffic issues in developing countries due to the rise in personal vehicle usage, leading to congestion, air pollution, and fuel wastage. Toll roads are introduced to collect taxes from users, and the paper highlights the need for efficient toll collection systems to address traffic problems.

The study evaluates various toll collection systems based on factors like time consumption, fuel wastage, traffic flow, payment modes, and processing efficiency. Conventional toll systems involve manual cash collection, leading to traffic jams, while automatic systems like FASTag and BookMyToll offer online payment options, reducing time and effort. The use of RFID technology in FASTag enables unique vehicle identification and automatic toll deduction.

The paper suggests that implementing automatic toll collection systems can save time, effort, and manpower, reducing traffic congestion at toll plazas. It proposes the use of Image Processing technology for toll collection, eliminating the need for physical devices at toll stations. Payment methods through mobile wallets, credit cards, or net banking are recommended for a seamless toll collection experience.

Overall, the study emphasizes the importance of adopting advanced toll collection systems to enhance traffic management, reduce fuel wastage, and provide a convenient payment experience for road users in India.

In another discussion on the integration of Vehicle-to-Everything (V2X) communication technology with highly automated and autonomous vehicles, Takacs and Haidegger [8] emphasize the crucial role V2X plays in enhancing both safety and operational efficiency. They identify several key challenges, including sustainability, scalability, cybersecurity, and interoperability, which must be addressed for successful implementation. Their work also highlights the assessment of V2X input data requirements at various autonomy levels as defined by SAE International, alongside regulatory challenges and scalability concerns in hybrid environments. Additionally, they explore the potential for IoT and AI-based information to impact non-automotive technical fields, further broadening the scope of V2X applications.

Dr. Maria Lenstrand [4] examines the human-centered design aspects of AI-driven voice assistants in autonomous vehicles, highlighting the importance of developing intuitive human-vehicle interfaces. The article emphasizes the need to enhance user experience while addressing critical issues such as safety, privacy, and regulatory challenges. Lenstrand also discusses the necessity for

advanced display and control systems that adapt dynamically to user interactions, particularly within the context of autonomous driving, where voice assistants play a pivotal role in facilitating communication and task management.

In another paper [3] focuses on the interaction between autonomous vehicles (AVs) and external road users (ERUs), stressing the importance of standardized communication protocols to enhance safety and trust. It identifies key information modes—state, intent, and response—that are critical for facilitating effective AV-ERU interactions. The authors emphasize significant gaps in current standards and the fragmented nature of existing approaches. They advocate for a collaborative, multi-stakeholder process to develop comprehensive guidelines tailored to diverse scenarios and user needs. The research ultimately seeks to ensure smoother AV integration into urban environments, fostering public acceptance and improving safety.

M. Sundarramurthi et al. [7] discusses the significant impact of the Internet of Things (IoT) on enhancing transportation systems, particularly in traffic management and operational efficiency. It explores various applications, such as automated toll collection using RFID technology and smart traffic signals that utilize image processing for vehicle monitoring and law enforcement. The authors emphasize the potential of IoT to streamline processes, reduce manpower, and improve safety and efficiency in transportation, while also suggesting further advancements and implementations to optimize these systems. Overall, the paper highlights the transformative role of IoT in modernizing the transportation industry.

V.Sandhya and A.Pravin [6] presented an Automatic Toll Gate Management and Vehicle Access Intelligent Control System utilizing an ARM7 microcontroller, designed to enhance the efficiency of toll collection and traffic management. The system comprises three main components: a vehicle unit equipped with an active RFID tag, a GSM modem, and a keypad; a tollgate unit with an RFID reader; and a central database system. When a vehicle approaches the toll gate, the RFID tag communicates with the tollgate unit, which sends the vehicle's identification information to the central database via ZigBee. If the vehicle's details and sufficient funds are verified, the toll gate opens automatically, allowing for seamless passage without stopping. This system aims to reduce congestion at toll booths and improve the overall toll collection process through automation and wireless communication technologies.

S Mitchell [5] discusses the importance of anticipating various issues related to machine-assisted tools in libraries, such as classifiers and extractors, which are expected to play a central role in generating standard metadata and extracting useful natural-language data over the next few years. It highlights the need for open models and development in the context of evolving research areas like named-entity identification and extraction, which focus on harvesting bibliographic metadata from resources. The text also addresses the tension between open access and restrictive information niches in libraries, emphasizing the necessity for libraries to rethink their organizational models and services to better align with the capabilities of the Internet and enhance resource visibility and accessibility.

Wang et al. [9] discuss the critical role of effective visualizations in intelligent vehicles to support human decision-making, emphasizing the efficiency of visual representation over traditional methods. Their research highlights how such visualizations can

assist various stakeholders, including vehicle engineers and drivers, in interpreting driving data and surroundings more effectively. They underscore the importance of accounting for individual differences in perception and personality when designing these interfaces, alongside the need for real-time adjustments to better cater to users. The study also explores the potential of emerging technologies like Augmented Reality (AR) and Virtual Reality (VR) in creating immersive experiences, suggesting future directions to further enhance vehicular visualizations and improve mobility-related decision-making.

Anderson et al. [1] examine the impact of center stack design on driver distraction, comparing touchscreen controls in modern vehicles to physical controls in legacy vehicles. Their research highlights that touchscreen interfaces, which lack tactile feedback, force drivers to divert their visual attention from the road, increasing distraction. Data from three naturalistic driving studies show that visual-manual tasks associated with touchscreen interactions are significantly more distracting than those involving physical controls. The findings underscore the heightened risk of distraction in modern vehicle designs, especially when combined with Level 2 driving automation, and stress the need for thoughtful human-machine interface design to enhance driver safety.

4 Approach

In response to the gap in existing research despite numerous studies, this study addresses our limited understanding of the human impact of AI and IoT adoption in developing countries, with a focus on the perspectives of toll collectors and user accessibility. We approach this inquiry through a mixed-methods framework, guided by grounded theory, social constructionism, and relativistic ontology. This involves fieldwork, in-depth interviews with toll collectors and users, and a comprehensive analysis of the collected data, which will be detailed in the following section.

4.1 Data Collection and Interviews

To thoroughly understand the human-centered aspects of AI and IoT adoption in Bangladesh's toll collection systems, we utilized a mixed-methods approach that integrates grounded theory, social constructionism, and relativistic ontology. This approach combines qualitative data from interviews with quantitative data on demographics and technology familiarity.

Interviews formed the backbone of our research. We conducted semi-structured interviews with ten toll collectors at a major toll plaza near Dhaka, exploring their perceptions of AI and IoT technologies in their workplace, their experiences with existing toll collection software, and any challenges they face. We also gathered demographic information such as age (Fig. 3), gender, educational background (Table. 1), work experience in toll collection (Table. 2, Fig. 4), familiarity with technology, and involvement in other occupations.

4.2 Data analysis

The participants in the study were all male smartphone users who handled their tasks without any formal training. Rather than receiving structured instruction, they acquired the necessary skills by learning from more experienced colleagues. Peer guidance played

a key role in helping them adapt to their work environment, where they managed their daily responsibilities during standard 8-hour shifts.

Over the course of their work, they frequently encountered situations that tested their judgment, particularly when dealing with potentially fraudulent currency. Torn or fake notes often passed through their hands, forcing them to make quick decisions under pressure. Despite these challenges, their day-to-day routine remained steady, defined by long hours and a reliance on their own experiences, as well as the shared knowledge of their peers.

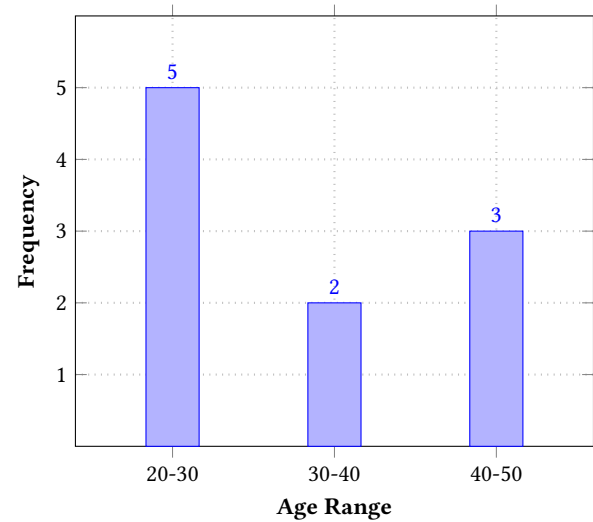


Figure 3: Age Distribution

Additionally, an in-depth interview with a toll plaza manager provided crucial insights into the managerial perspective on the implementation of AI and IoT. Understanding the management perspective complements the experiences of toll collectors and contributes to a more complete picture.

Table 1: Educational background

Education Level	Number of Operators
Undergraduate (Bachelor degree)	3
Graduate (Masters degree)	1
Diploma	2
Higher school	4

To explore user experiences and potential accessibility concerns, we also conducted interviews with a sample of bus and truck drivers who frequently use the toll booths.

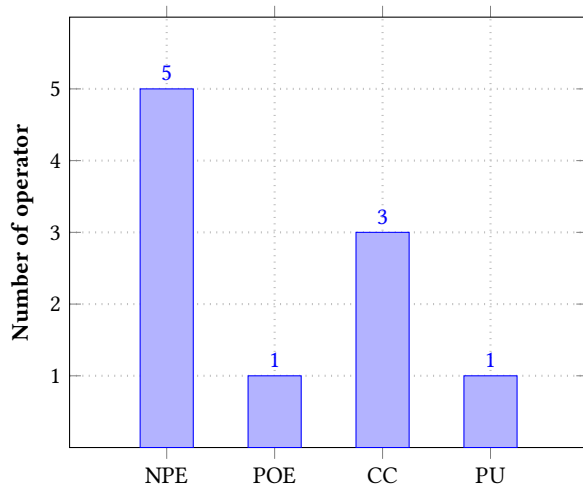
Quantitative data was collected concurrently with the interviews, focusing on participants' demographics and technology familiarity. This data helps construct a profile of the toll collector population and user base, which we will use for descriptive analysis.

By combining qualitative and quantitative data collection methods within a framework grounded in theories of grounded theory,

Table 2: Experience with Toll Plaza

Participants	Experience (Years)	Experience (Number of Toll Plazas)
P1	3	4
P2	10	5
P3	1	1
P4	1	1
P5	4	4
P6	10	4
P7	2	3
P8	1	1
P9	2	1
P10	1	1

social constructionism, and relativistic ontology, we captured the nuanced subjective experiences and perspectives of toll collectors and users, along with objective data on their backgrounds. This comprehensive approach will be further discussed in the next section, outlining the data analysis methods used to draw meaningful insights from the collected information.

**Figure 4: Experience with Computers**

NPE - No Previous Experience
POE - Previous Office Experience
CC - Course on Computer
PU - Personal User

5 Findings

For the analysis of descriptive answers, we employed thematic analysis to evaluate the qualitative data. During the interview process, we aimed to create a comfortable environment for the toll collectors to elicit authentic responses. To facilitate effective communication, we made an effort to engage with them in their native language or dialect whenever possible.

The interviews with 10 toll booth operators and 1 toll plaza manager offered insights into the research questions regarding the impact of AI and IoT-based toll collection systems on the transportation system, as well as their accessibility for people with varying levels of education. Below are the key themes identified from the interviews, along with excerpts supporting each theme:

5.1 Efficiency and Streamlining vs. Complexity and Challenges

AI and IoT-based toll collection systems are designed to enhance efficiency and streamline the toll collection process. However, operators reported facing various complexities. For instance, inaccuracies in AI-based vehicle detection sometimes lead to misclassification of vehicle types, complicating the toll collection process. Additionally, operators found challenges with the boom barrier's automation, as it could open prematurely before verifying the authenticity of cash transactions, such as torn or fake currency. Manual interventions were sometimes necessary, but these could slow down operations and cause frustration for both operators and drivers.

Excerpts:

- "Using mouse for selecting vehicle type is tiresome."
- "Auto selection of vehicle type is wrong in significant time."
- "If we manually control the barrier, it is time consuming and tiresome."

These excerpts highlight the need for improved AI and IoT technologies that reduce errors and streamline operations.

5.2 Education and Adaptability to AI and IoT Technologies

The operators' levels of education influenced their adaptability to AI and IoT-powered toll collection systems. Those with higher education levels or more experience were comfortable with advanced AI and autofill technologies and preferred streamlined processes. Conversely, newer operators or those with lower education levels found the current processes acceptable but desired more user-friendly and customized solutions, such as specialized keyboards, to increase efficiency and reduce errors.

Excerpts:

- "Experienced toll collectors said, 'Obviously these softwares are easing the task.'"
- "Newer toll collector said, 'They are okay with the current process.'"
- "They all want customized keyboard for faster work flow."

This indicates that the design and implementation of toll collection systems should consider the varying technological literacy and preferences of operators to ensure ease of use and efficiency across all levels.

5.3 Payment Preferences and Fraud Detection

Operators expressed a strong preference for cash payments due to their speed and ease of use, even though this posed risks such as

handling torn or fake currency. Online payment methods were generally seen as slow and prone to errors, making them less favorable for toll collection. The ability to detect fraud, such as identifying counterfeit currency, varied depending on the operator's experience and education level.

Excerpts:

- "All of them are strongly against any kind of online payment."
- "Experienced people easily detect these kind of cases."
- "We need some better solution here."

This preference for cash payments and concerns about fraud detection suggest that future toll collection systems should address the challenges associated with these aspects, potentially through improved training or technological solutions.

In summary, the findings underscore the need to balance the advantages of AI and IoT-based toll collection systems with the complexities and challenges they introduce. It also emphasizes the importance of considering the varying educational backgrounds and experiences of toll booth operators to ensure that the systems are accessible and efficient for all. Future work should aim to address these concerns through user-centered design and technological innovations that can improve overall toll booth operations.

6 Discussion

The development of the ANPR (Automatic Number Plate Recognition) system and the one-page software directly addresses the needs and requirements of the toll booth operators as highlighted in the interview analysis. These technological solutions demonstrate progress toward improving efficiency and streamlining toll booth operations.

The ANPR system's reported accuracy of approximately 95% is a promising outcome. Operators' feedback indicates that this high level of accuracy in recognizing vehicle number plates can significantly speed up the toll collection process, particularly when paired with the ability to store data in a local database for frequently recurring vehicles. This aligns with the operators' desire for streamlined, quick processing, as one operator mentioned, *"These data should be stored in a local database to fasten the process for frequently moving vehicles' transactions."*

However, the operators' hesitance toward completely automating the toll collection process is noteworthy. They emphasized that while AI and IoT technologies can aid in the identification of vehicles, they are not yet capable of managing all toll collection scenarios autonomously. The operators' experience and judgment are essential in addressing vehicles that may not conform to standard classifications or require special tolling due to their deformed structure. They commented, *"Many vehicles do not follow proper rules and regulation. Due to their deformed structure, the toll amount cannot be detected only by the series number. Experienced toll collectors should see the situation and solve toll amount."*

The one-page software aligns with the operators' desire for a streamlined user interface and less interaction with software systems. The simplicity and efficiency of the software have been well-received by the operators, who expressed satisfaction with the design. Nonetheless, they indicated the need to test the software in a real-world setting before providing comprehensive feedback. As

one operator noted, *"They seemed to be satisfied with the software, but are willing to comment after using it, as only the UI was provided."*

In summary, the ANPR system and one-page software offer substantial benefits in terms of streamlining toll booth operations and enhancing efficiency. However, the operators' feedback emphasizes the importance of maintaining human oversight in the toll collection process to address complex or irregular situations that AI and IoT systems may struggle to manage independently. Moving forward, future developments should focus on optimizing the balance between technology and human expertise, ensuring that the system remains efficient while accommodating the nuanced judgment required in specific toll collection cases.

7 Conclusion

The research has demonstrated the potential benefits of AI and IoT-based toll collection systems, particularly the ANPR system and one-page software, in enhancing efficiency and streamlining toll booth operations. The high accuracy of the ANPR system (95%) supports faster and more efficient processing of toll transactions, especially when combined with local data storage for frequently recurring vehicles. The one-page software's user-friendly interface addresses operators' needs for a straightforward and efficient system.

However, the interviews also highlighted the necessity of retaining human oversight in toll collection due to the complexities that arise in real-world scenarios, such as vehicles with non-standard classifications or deformed structures. Experienced toll collectors play a vital role in these situations, ensuring accurate toll assessments and addressing unique circumstances that AI and IoT systems cannot manage autonomously.

In conclusion, the combination of AI and IoT technologies with human expertise can lead to significant improvements in toll booth operations, offering a balance between efficiency and adaptability.

Future work in this area should focus on the following aspects:

Explore ways to optimize the collaboration between AI systems and human operators. Develop interfaces and systems that allow operators to intervene seamlessly when necessary, providing the flexibility to handle non-standard tolling scenarios effectively.

Conduct in-depth user testing of the one-page software to gather detailed feedback from toll booth operators. This feedback will be invaluable in refining the software and ensuring it meets their practical needs.

Investigate ways to improve the detection and handling of fraudulent activities, such as torn or fake currency, within the toll collection process. This could involve integrating advanced fraud detection technologies or providing additional training for operators.

Continue exploring customizable solutions, such as specialized keyboards or user interface options, to cater to operators with varying levels of education and experience. These adjustments can enhance workflow efficiency and reduce errors.

Analyze the balance between automated decisions and manual interventions in toll collection. Determine the scenarios in which human judgment is essential and where AI can be most effective.

By focusing on these areas, future research and development can contribute to the evolution of toll collection systems that are not only efficient and technologically advanced but also flexible

enough to accommodate the nuanced judgment and expertise of human operators. This approach will ultimately lead to improved toll collection processes and a better experience for both operators and road users.

Acknowledgments

Thanks to Spectrum Engineering Consortium (Pvt.) Ltd. for providing the opportunity to conduct the interviews.

References

- [1] Gabriel T Anderson, Jacobo Antona-Makoshi, and Charlie Klauer. 2024. *Human-Machine Interface Review: A Comparison of Legacy and Touch-Based Center Stack Controls*. Technical Report. National Surface Transportation Safety Center for Excellence.
- [2] Bharavi Joshi, Kajal Bhagat, Hetakshi Desai, Malvik Patel, and Jekishan Kantilal Parmar. 2017. A comparative study of Toll Collection Systems in India. *Int. J. Eng. Res. Dev* 13, 11 (2017), 68–71.
- [3] Swapna Joshi, Avram Block, and Paul Schmitt. 2023. Autonomous Vehicle and External Road User Interfaces: Mapping of Standards Gaps and Opportunities. In *Adjunct Proceedings of the 15th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. 30–35.
- [4] Maria Lenstrand. 2023. Human-Centered Design of AI-driven Voice Assistants for Autonomous Vehicle Interactions. *Journal of Bioinformatics and Artificial Intelligence* 3, 2 (2023), 37–55.
- [5] Steve Mitchell. 2006. Machine Assistance in Collection Building: New Tools, Research, Issues and Reflections. *Information Technology and Libraries* 25, 4 (2006), 190–216.
- [6] V Sandhya and A Pravin. 2012. Automatic Toll Gate Management and Vehicle Access Intelligent Control System Based on ARM7 Microcontroller. *International Journal of Engineering Research and Technology* 1, 5 (2012), 2278–0181.
- [7] M Sundarramurthi, Aravind R Krishnan, S Nandita, and Kruthi Chandrashekar. [n. d.]. Emerging IoT Technologies in Transportation. ([n. d.]).
- [8] Arpad Takacs and Tamas Haidegger. 2024. A Method for Mapping V2X Communication Requirements to Highly Automated and Autonomous Vehicle Functions. *Future Internet* 16, 4 (2024), 108.
- [9] Xumeng Wang, Xingxia Wang, Siji Ma, Wei Chen, and Fei-Yue Wang. 2023. Vehicular Visualization: Enhancing Mobility Decisions With Human–Machine Interactions. *IEEE Transactions on Intelligent Vehicles* 8, 11 (2023), 4653–4663.
- [10] Khorshadul Haque Zoy, Mahir Shahrier, and Armana Sabiha Huq. 2020. A systematic review of electronic toll collection systems. In *International Conference on Transportation Research*.